



Precision Cardiovascular Risk Intelligence Platform

Dashboard Overview & Feature Guide

A comprehensive architectural and functional walkthrough of the multi-modal cardiovascular risk simulation workstation, designed for cardiologists, academic supervisors, and technical reviewers.

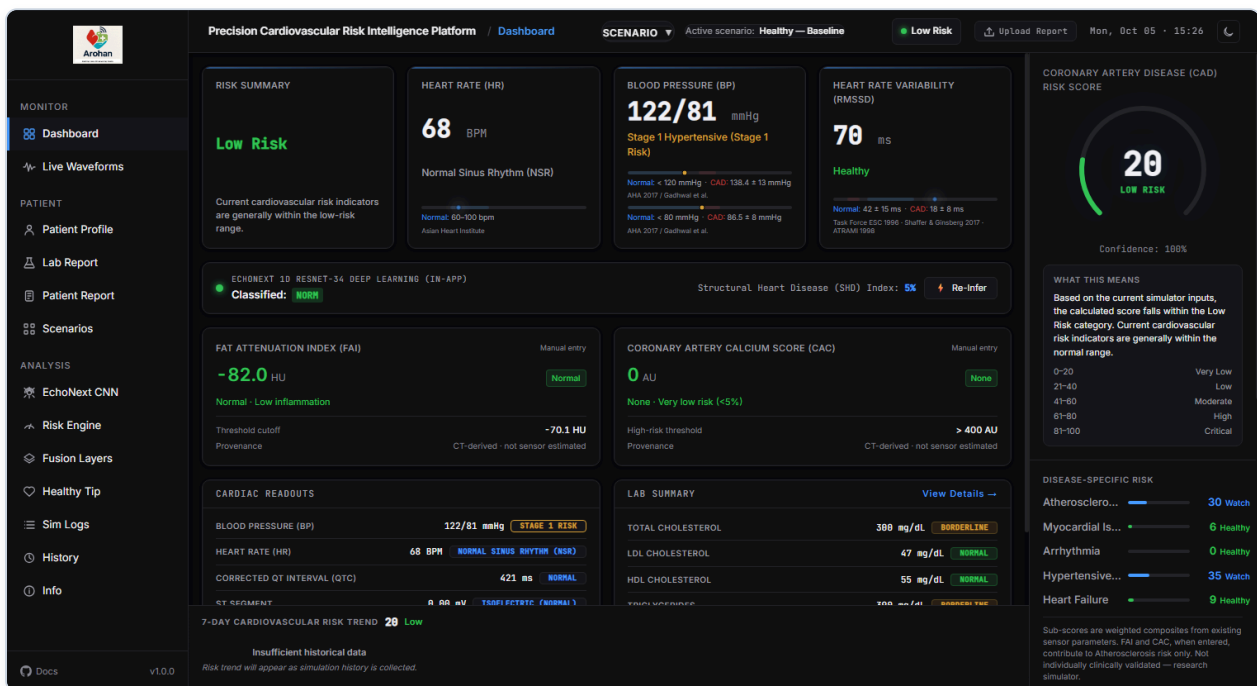


Figure 1.0: Real-time clinical workstation interface displaying multi-parameter telemetry, risk scoring, and longitudinal trends.

Research and Educational Prototype Notice: This software is a simulation, modeling, and clinical decision-support workstation created for exploratory research and educational evaluation. It is not an FDA/CE cleared medical device and does not substitute for independent clinical judgment.

TARGET AUDIENCE

Supervisors, Reviewers & Clinicians

IMPLEMENTATION

Web-Based Telemetry Workstation

VERSION & DATE

v1.0.0 · October 2026

What the Platform Does

Purpose, Context, and End-to-End Processing Pipeline

The Precision Cardiovascular Risk Intelligence Platform was built to address a fundamental challenge in preventive cardiology: how to synthesize diverse, asynchronous patient data into a coherent, actionable risk assessment.

Core Objective & Clinical Context

Modern cardiovascular care relies on fragmented data sources—wearable vitals, laboratory blood tests, electrocardiography, and advanced coronary computed tomography. This platform acts as a unified clinical intelligence hub, enabling clinicians and researchers to test physiological scenarios, observe simulated biomarker interactions, and inspect how individual variables drive composite coronary disease risk.

End-to-End Data Pipeline Flow

1

Patient & Scenario Inputs

Active demographic profile, pathological presets, or ingested clinical laboratory documents.

2

Physiological Parameter Synthesis

Generates synchronized hemodynamic and electrophysiological values (HR, BP, HRV, ECG timing).

3

Signal & Sensor Processing

EchoNext 1D ResNet-34 deep learning inference analyzes raw rhythm morphology and structural indices.

4

Multi-Factor Risk Assessment

Weighted clinical risk engine integrates Framingham, Reynolds, SCORE2, and biomarker heuristics.

5

Interactive Dashboard & Patient Report

Live visual workstation updates synchronously and generates exportable clinical PDF reports.

Summary: By simulating the full continuum from raw pulse-wave telemetry to high-level clinical reports, the platform provides a transparent sandbox for exploring coronary artery disease trajectories.

Dashboard at a Glance

Layout Structure and Primary Workstation Zones

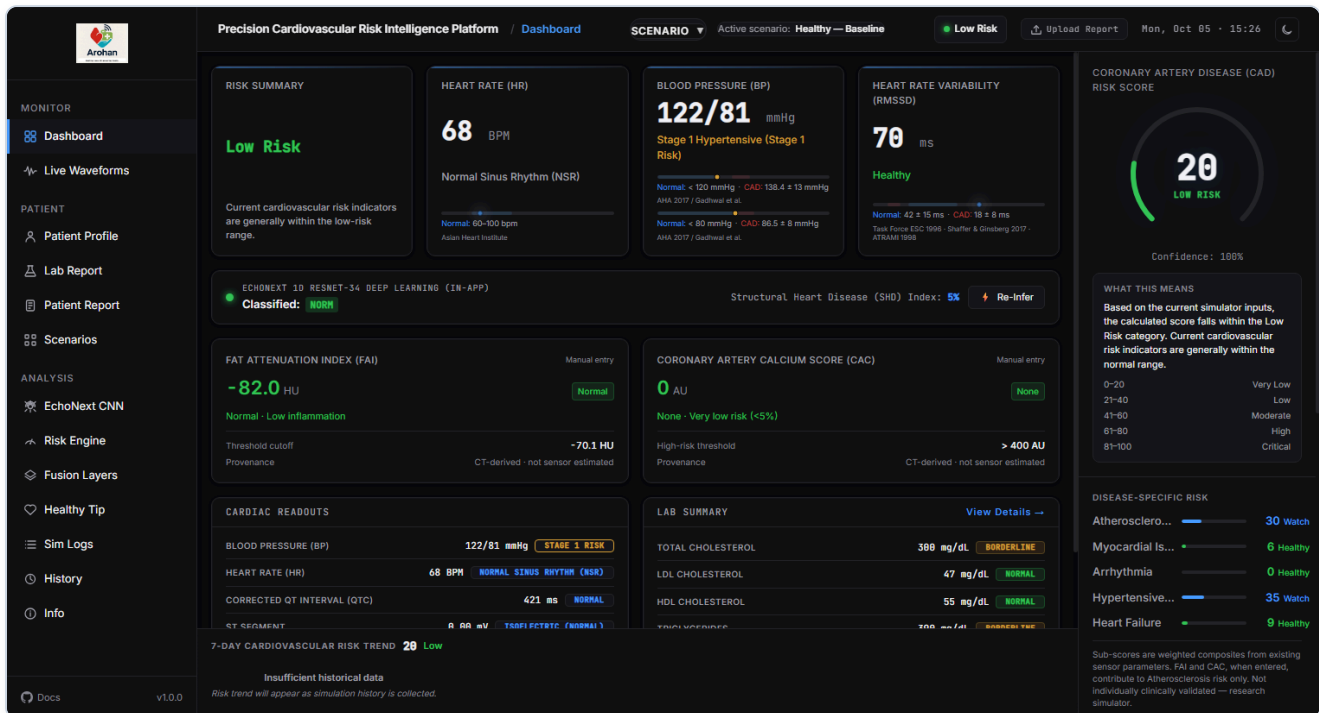


Figure 3.1: Primary workstation interface divided into persistent navigation, active telemetry, and risk assessment panels.

- 1 Application Header:** Uncontained Arohan branding, platform title, active breadcrumb, status badge, and live UTC clock.
- 2 Scenario Selector:** Dropdown to toggle simulated physiological profiles (Healthy, CAD, CVD states).
- 3 Current Risk Score:** 0–100 composite coronary artery disease risk gauge with color-coded risk band.
- 4 Heart Rate (HR):** Beats per minute with sinus rhythm classification and physiological reference range.
- 5 Blood Pressure (BP):** Systolic/diastolic arterial pressures classified by JNC 8 / AHA guidelines.
- 6 Heart Rate Variability (RMSSD):** Beat-to-beat root mean square variance indicating autonomic balance.
- 7 ECG Model Status:** In-app EchoNext 1D ResNet-34 classifier detecting rhythm abnormalities and structural heart index.
- 8 Fat Attenuation Index (FAI):** Perivascular adipose tissue radiodensity measuring coronary inflammation.
- 9 Coronary Calcium Score (CAC):** Agatston calcification burden indicating anatomical plaque presence.
- 10 Cardiac Readouts:** Electrophysiological timing parameters including QTc interval, ST deviation, PTT, and SpO2.
- 11 Lab Summary:** Lipid panel indicators (Total Cholesterol, LDL, HDL, Triglycerides) with clinical threshold flags.
- 12 Disease-Specific Risk:** Sub-risk bar indicators for Atherosclerosis, Ischemia, Arrhythmia, and Heart Failure.
- 13 Contributions Ranking:** Ranked feature influence quantifying parameter weight on the current score.
- 14 7-Day Risk Trend:** Recorded longitudinal chart tracing historical risk score stability and trajectory.

Scenario System

Controlled Pathological States for Research and Demonstration

The platform includes a dedicated scenario engine that dynamically reconfigures simulated inputs to reflect distinct cardiovascular conditions. This allows reviewers to observe how the risk engine responds to acute and chronic disease phenotypes.

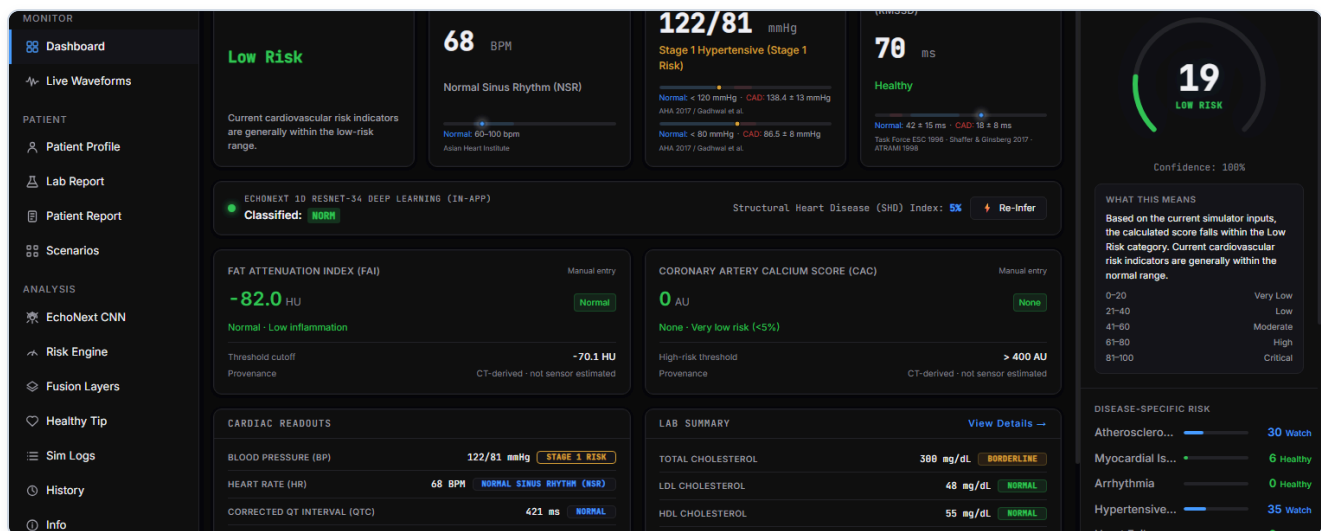


Figure 4.1: Scenario selection dropdown showing pre-configured physiological archetypes across three core categories.

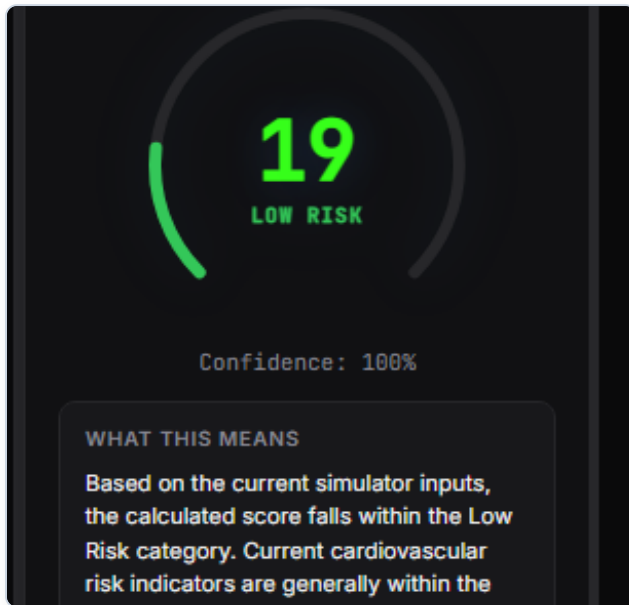
Category	Preset Archetypes	Physiological Effect on Simulator
Healthy	Baseline, Post-Exercise, Athletic Profile	Maintains normal sinus rhythm, optimal BP (<120/80 mmHg), healthy HRV (>50 ms), and negative inflammatory markers.
Coronary Artery Disease (CAD)	Early Plaque, Significant Stenosis, Unstable Plaque	Elevates FAI (-65 HU), increases CAC score (>200 AU), elevates LDL, and induces subtle ST-segment depressions during stress.
Cardiovascular Disease (CVD)	Stage 1 & 2 Hypertension, Arrhythmia / AFib, Diabetic Dyslipidemia, Heart Failure	Induces sustained arterial hypertension (>140/90 mmHg), depresses HRV (<20 ms), triggers QTc prolongation, and elevates myocardial ischemia sub-scores.

Important Simulation Distinction: Scenarios are reproducible mathematical models designed to test risk algorithms and UI responsiveness. They do not represent real-time telemetry from living patients.

Primary Output: Risk Score

Composite Multi-Modal Cardiovascular Risk Assessment

The primary output of the platform is the Coronary Artery Disease (CAD) Risk Score, displayed prominently in the right-hand inspection panel as a normalized radial gauge ranging from 0 to 100.



What the Score Represents

Unlike single-variable calculators (e.g. lipid-only charts), this score synthesizes autonomic function (HRV), hemodynamic load (BP), electrophysiology (ECG), CT biomarkers (FAI/CAC), and circulating lipids into one holistic metric.

Confidence Metric

The gauge displays an explicit confidence score (0–100%) that dynamically degrades if simulated sensor motion artifacts or missing lab values compromise input signal integrity.

Risk Band Stratification & Clinical Meaning

Range	Band	Clinical Interpretation in Simulator
0 – 20	Very Low	All biomarkers, hemodynamics, and lipid levels reside strictly within normative baselines.
21 – 40	Low Risk	Mild isolated elevation (e.g. pre-hypertension) with normal vascular imaging and lipid targets.
41 – 60	Moderate	Borderline dyslipidemia combined with Stage 1 hypertension or mild calcification burden (CAC 1–99).
61 – 80	High Risk	Multiple significant risk drivers present: Stage 2 hypertension, active perivascular inflammation (FAI > -70 HU).
81 – 100	Critical	Severe multi-system risk: heavy coronary calcification (CAC > 400), ischemic ECG changes, severe dyslipidemia.

Research Disclaimer: The displayed score is generated by the simulator's internal assessment logic and should be interpreted as a research and educational output rather than a validated clinical diagnostic score.

7-Day Risk Trend

Tracking Historical Trajectories and Treatment Responses

Cardiovascular risk is not static; single-point snapshots often conceal emerging destabilization or positive treatment response. The platform features a persistent 7-day risk trend bar anchored at the bottom of the central monitor.

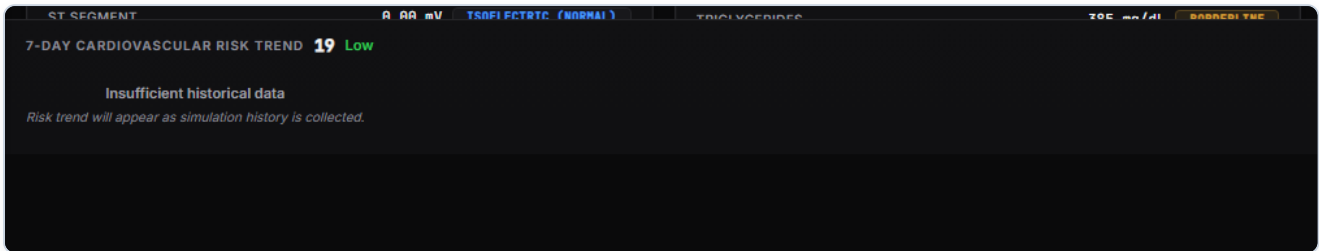


Figure 6.1: Longitudinal risk trend showing 7 recorded daily points (D1 through D7), current score (19 Low), and a stable trajectory indicator.

Why Historical Movement Matters

In clinical cardiology, a patient whose risk score is 35 and steadily declining represents a very different prognosis from a patient whose score was 15 three days ago and has climbed sharply to 35. Longitudinal tracking allows clinicians to evaluate whether therapeutic interventions (e.g. statin initiation or antihypertensive titration) are successfully stabilizing vascular health.

Trajectory Pattern	Visual Representation	Clinical Significance in Simulator
Stable (→ Stable)	Horizontal green curve (± 2 points variance)	Indicates steady physiological homeostasis without acute progressive plaque activity or hemodynamic volatility.
Increasing (↗ Rising)	Upward slope transitioning into amber/red	Reflects progressive worsening in one or more drivers (e.g., escalating blood pressure, missed medication, or rising FAI).
Decreasing (↘ Improving)	Downward curve trending toward low-risk band	Simulates positive therapeutic response to lifestyle optimization, lipid-lowering therapy, or blood pressure control.

History Data Integrity: The chart reflects actual logged simulation history. If a new session has fewer than 7 days recorded, the system displays an "Insufficient historical data" indicator rather than fabricating artificial historical points.

Physiological Parameters

Real-Time Autonomic and Hemodynamic Telemetry Readouts

The upper telemetry grid displays four real-time parameter cards that update continuously based on the active sensor simulation pipeline.

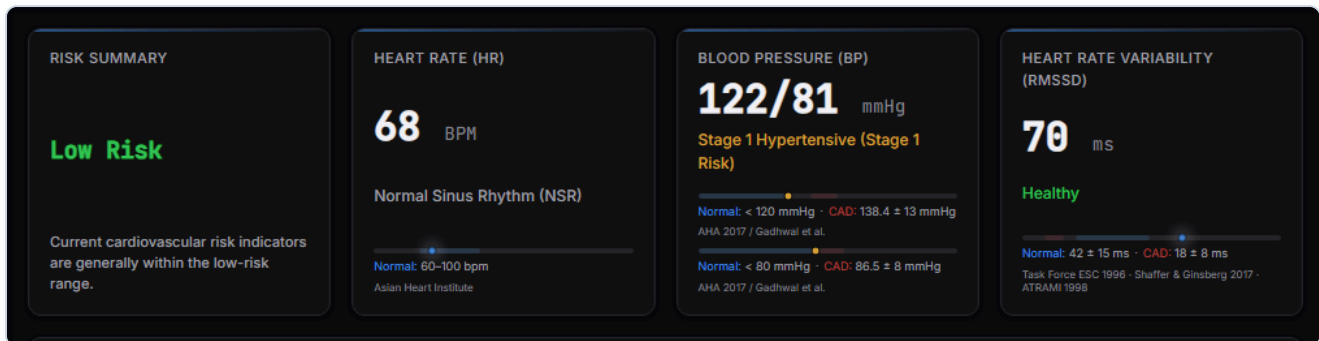


Figure 7.1: Continuous physiological readouts: Risk Summary, Heart Rate, Blood Pressure, and Heart Rate Variability.

1. Heart Rate (HR) Target: 60 – 100 BPM

Represents the frequency of ventricular contractions per minute, extracted from simulated ECG and photoplethysmography (PPG) waveforms. The card indicates baseline sinus rhythm and alerts on resting bradycardia (<50 bpm) or resting tachycardia (>100 bpm).

2. Blood Pressure (BP) AHA / JNC 8 Standard

Measures peak systolic arterial pressure during contraction and minimum diastolic pressure during filling (mmHg). The system provides dual range-indicator bars and classifies readings into Normal (<120/80), Elevated (120–129/<80), Stage 1 Hypertension (130–139/80–89), or Stage 2 Hypertension (≥140/≥90 mmHg).

3. Heart Rate Variability (RMSSD) Normal: 42 ± 15 ms

Root Mean Square of Successive RR interval Differences (RMSSD) quantifies beat-to-beat variability in milliseconds. Higher values reflect robust parasympathetic vagal tone, while depressed HRV (<20 ms) indicates sympathetic overdrive and heightened cardiovascular vulnerability.

4. Cardiac Electrophysiological Readouts QTc, ST, PTT, SpO2

Secondary readouts include Bazett-corrected QT interval (QTc, normal <440 ms in men, <460 ms in women), ST-segment deviation (-0.05 to +0.05 mV), Pulse Transit Time (PTT), and peripheral blood oxygen saturation (SpO2 >95%).

Fat Attenuation Index (FAI) & CAC

CT-Derived Molecular Inflammation and Plaque Burden Markers

Conventional risk tools rely heavily on systemic biomarkers. The platform integrates two advanced imaging parameters that reflect coronary biological activity and anatomical disease.

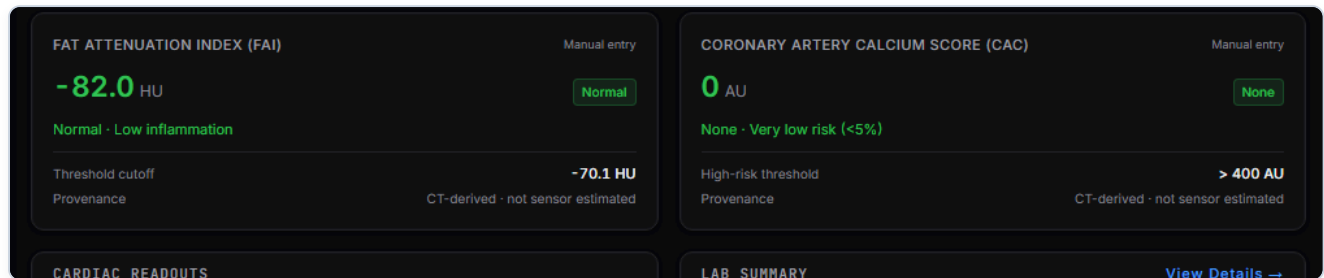


Figure 8.1: Specialized biomarker cards for perivascular Fat Attenuation Index (left) and Coronary Artery Calcium Score (right).

Fat Attenuation Index (FAI)

Perivascular Adipose Tissue (PVAT) Radiodensity

What it measures: Derived from coronary computed tomography angiography (CCTA). When coronary arteries become inflamed, cytokines diffuse into surrounding fat, preventing adipocyte lipid accumulation and shifting CT density toward water.

Threshold: Measured in Hounsfield Units (HU). Values more negative than **-70.1 HU** are normal. Values between -70.0 and -60.0 HU indicate elevated perivascular inflammation.

Significance: Identifies vulnerable, actively inflamed plaques *before* luminal narrowing or structural rupture occurs.

Coronary Calcium Score (CAC)

Agatston Plaque Burden Quantification

What it measures: Quantifies calcium deposits within the coronary arterial intima on non-contrast chest CT.

Agatston Categories:

- **0 AU:** None (Very low 5-year event risk <1%)
- **1 – 99 AU:** Mild plaque burden
- **100 – 399 AU:** Moderate plaque burden
- **≥ 400 AU:** Extensive calcification (High risk)

Significance: A definitive physical marker of established atherosclerosis that reclassifies borderline clinical risk categories.

Clinical Note: In this simulator, FAI and CAC are manually entered or ingested from imaging report uploads, as they cannot be estimated directly from non-invasive skin/pulse sensors.

Laboratory Lipid Summary

Circulating Metabolic Biomarkers and Atherogenic Lipoproteins

Lipid management is central to atherosclerotic prevention. The platform displays a consolidated Lab Summary card that tracks primary circulating lipoprotein fractions against standard clinical cutoffs.

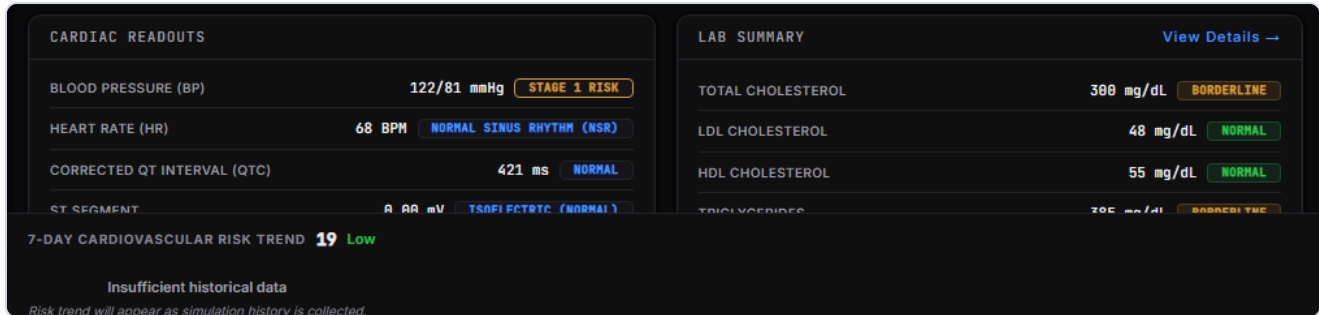


Figure 9.1: Lower dashboard card displaying cardiac telemetry readouts alongside the active serum lipid profile.

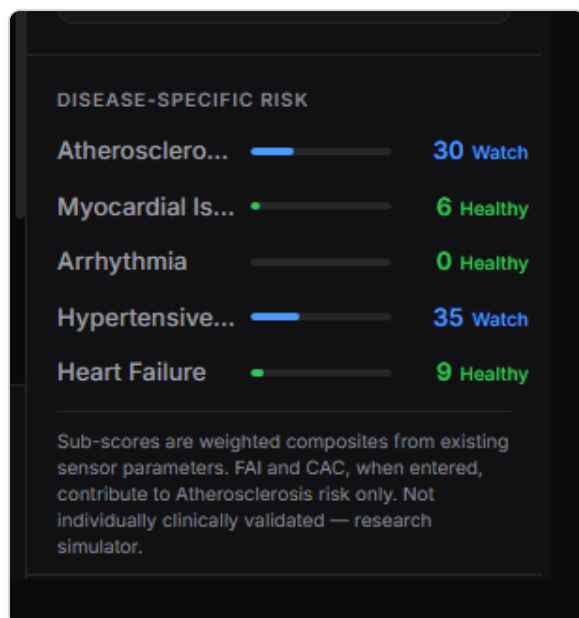
Lipid Fraction	Clinical Target	Relevance to Cardiovascular Risk in Model
Total Cholesterol (TC)	< 200 mg/dL	Overall sum of circulating cholesterol particles. Values >240 mg/dL contribute substantial baseline points to the composite risk engine.
LDL Cholesterol	< 100 mg/dL (<70 in high-risk)	Low-Density Lipoprotein is the principal atherogenic particle responsible for subendothelial cholesterol retention and fatty streak formation.
HDL Cholesterol	> 40 mg/dL (M) > 50 mg/dL (F)	High-Density Lipoprotein mediates reverse cholesterol transport back to the liver. In the simulator, high HDL acts as a protective negative weight.
Triglycerides (TG)	< 150 mg/dL	Circulating neutral fats. Values >200 mg/dL signify atherogenic remnant lipoproteins and insulin resistance, elevating metabolic sub-scores.
Apolipoprotein B (ApoB)	< 80 mg/dL	Direct count of all atherogenic particles (LDL, VLDL, IDL). Often provides superior predictive value compared to calculated LDL in dyslipidemia.

Data Ingestion: Lipid parameters can be populated via manual profile inputs, automated scenario presets, or parsed directly from uploaded lab PDF reports using the integrated OCR engine.

Disease-Specific Risk Sub-Scores

Differential Risk Decomposition Across Cardiovascular Pathologies

Coronary disease rarely exists in isolation. To provide nuanced decision support, the platform breaks down the patient's global risk into five disease-specific sub-scores displayed in the right inspection panel.



Each sub-score is an independently calculated 0–100 index with status tags (**Healthy**, **Watch**, or **Elevated**), allowing clinicians to pinpoint whether a patient's primary vulnerability is ischemic, electrical, or vascular.

1. Atherosclerosis (0–100)

Driven predominantly by anatomical calcification (CAC), active perivascular fat inflammation (FAI), and circulating LDL/ApoB particles.

2. Myocardial Ischemia (0–100)

Calculated from ST-segment depression magnitude, Rate-Pressure Product ($RPP = HR \times SBP$), and coronary stenosis estimates.

3. Arrhythmia & Conduction (0–100)

Driven by QTc interval prolongation, autonomic HRV depression, and rhythm irregularity flags from the EchoNext deep learning model.

4. Hypertensive Heart Disease

Reflects chronic afterload strain derived from sustained systolic elevation (> 140 mmHg), diastolic pressure, and wide pulse pressure.

5. Heart Failure Trajectory

Early risk signal driven by depressed HRV, resting tachycardia, elevated blood pressure, and EchoNext structural heart disease (SHD) index.

Statistical Notice: These sub-scores represent simulator-derived relative indices rather than epidemiologically validated actuarial event probabilities.

Risk Contributions & Attribution

Transparent Factor Weighting and Feature Attribution

A recognized barrier to adopting clinical decision-support algorithms is the "black-box" problem. The platform incorporates transparent Explainable AI (XAI) feature attribution, allowing users to see exactly what parameters drive the overall score.

What is Driving the Current Assessment?

The Contributions module deconstructs the composite risk score into proportional weighted points. Rather than delivering an unexplained number (e.g. 58), the system isolates whether the score is being pushed upward by Blood Pressure, Calcium Score, or LDL levels.

Risk Driver Parameter	Typical Influence Weight	Simulator Attribution Logic
Coronary Calcium (CAC)	20 – 30% Weight	Highest single weight in established disease; scales logarithmically with Agatston AU score.
Blood Pressure (Systolic)	18 – 25% Weight	Calculates excess pressure above 120 mmHg; non-linear acceleration when SBP exceeds 140 mmHg.
LDL Cholesterol / ApoB	15 – 22% Weight	Direct atherogenic particle penalty proportional to circulating concentration above 100 mg/dL.
Fat Attenuation Index	12 – 18% Weight	Perivascular inflammation multiplier that amplifies plaque vulnerability scores.
Autonomic State (HRV)	8 – 14% Weight	Inverse penalty: low RMSSD (<25 ms) adds risk points; high RMSSD (>50 ms) applies a protective credit.
ECG Morphology / EchoNext	6 – 12% Weight	Penalties triggered by ST depression, prolonged QTc intervals, or structural heart disease flags.

Clinical Utility of Attribution

By identifying the dominant contributor for a given patient profile, the decision-support system suggests personalized therapeutic targets—for instance, indicating that lowering SBP by 15 mmHg would yield a greater risk reduction than further intensifying statin therapy.

Attribution Caveat: Factor contributions reflect the simulator's internal mathematical weighting rules and should not be construed as independently proven causal relationships in individual patients.

Patient Report Page

Consolidated Clinical Summary and Export Documentation

The Patient Report module transforms live simulator telemetry and algorithmic inferences into a structured, audit-ready clinical document suitable for multidisciplinary cardiology reviews.

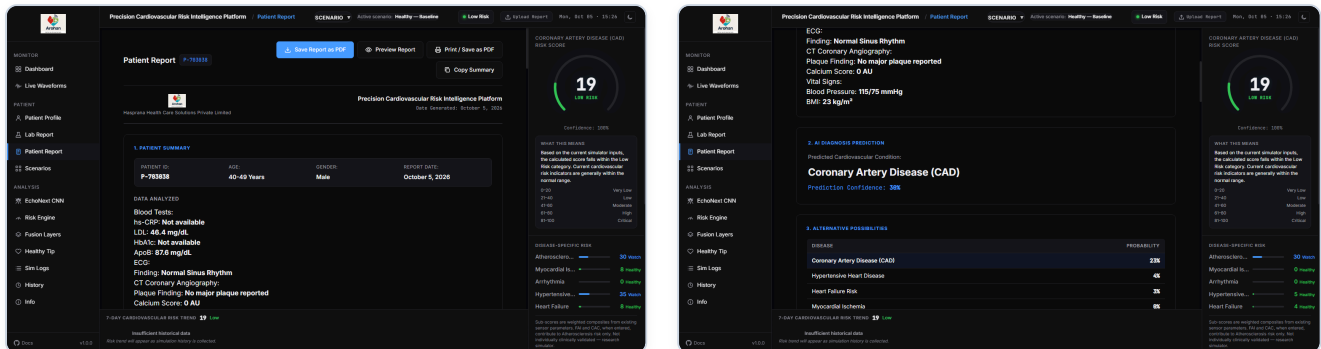


Figure 12.1: Patient Report layout showing executive summary, demographics, risk stratification, and XAI recommendations.

1. Patient Profile & Demographics

Documents age, sex, smoking history, diabetic status, BMI, and baseline cardiovascular medical history.

2. Stratified Risk & Trajectory

Consolidates the composite risk score, risk band, confidence rating, and longitudinal 7-day trend summary.

3. Multi-System Biomarker Summary

Tabulates hemodynamic vitals, full serum lipid panel, CT imaging indices (FAI/CAC), and ECG metrics.

4. Evidence-Based Recommendations

Generates prioritized pharmacological and lifestyle recommendations targeted directly to the patient's dominant risk factors.

Export Functionality: The Patient Report includes an automated PDF export engine (jsPDF) that compiles the entire report into a downloadable clinical summary document.

Report Upload & Document OCR

Client-Side Clinical Document Parsing and Ingestion Pipeline

Entering laboratory and imaging values manually can be tedious and prone to transcription errors. The platform features an integrated Report Upload slide-in panel that ingests external clinical documents.

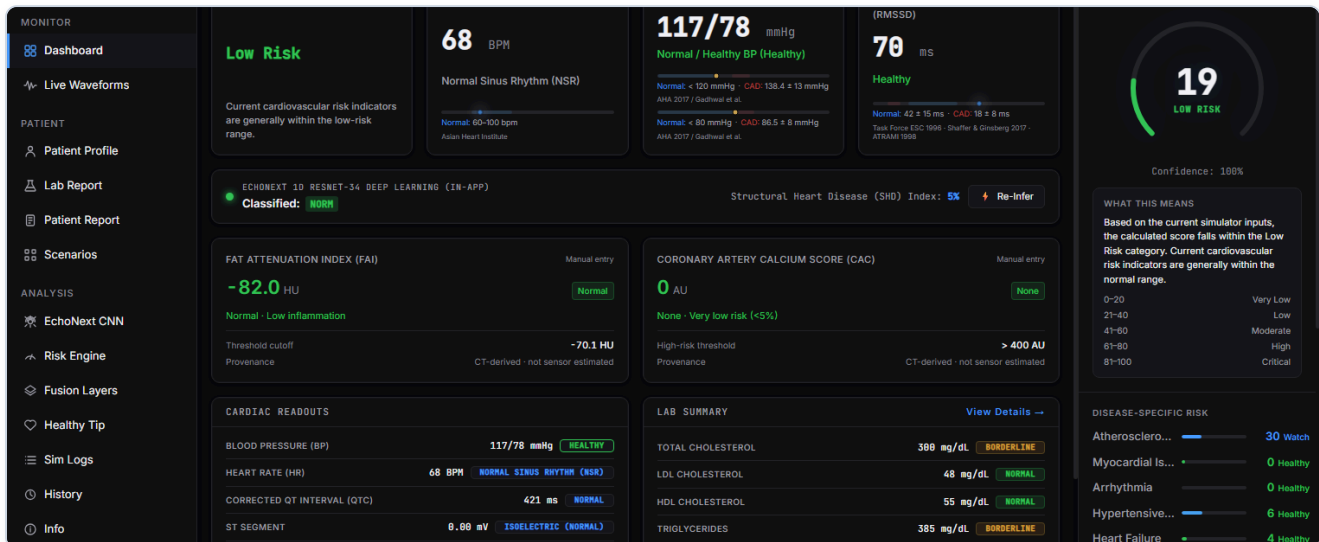


Figure 13.1: Slide-in Report Upload panel showing drag-and-drop zone, supported document formats, and extracted parameter review.

Document Ingestion Pipeline

- A Supported File Types:** Accepts standard laboratory reports in PDF, PNG, or JPEG formats up to 10 MB.
- B Client-Side OCR Processing:** Uses in-browser Tesseract.js optical character recognition to extract numeric values without sending PHI to external servers.
- C Extracted Target Fields:** Automatically identifies Total Cholesterol, LDL, HDL, Triglycerides, Blood Pressure, and CAC scores.
- D Human-in-the-Loop Review:** Displays extracted values in an editable preview grid, requiring clinician confirmation before applying them to the active simulator state.

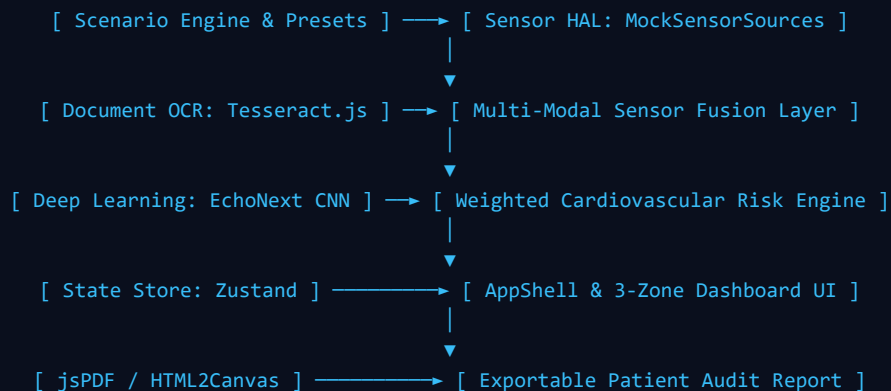
Data Source Category	Origin in Platform	Update Frequency & Mutability
Uploaded Data	Parsed clinical lab documents	Static until new document is uploaded; overrides baseline simulator inputs upon user confirmation.
Simulated Data	Internal telemetry generators	Continuously streaming; updates HR, BP fluctuations, and waveform traces in real time.
Manual Data	User profile & modal entry	Direct clinician override via patient profile modal or dedicated biomarker inputs.

Technical Architecture & Stack

Engineering Framework, Component Hierarchy, and Dependencies

The application is engineered as a modern, high-performance web workstation using React 19, TypeScript, and Vite. The codebase enforces strict separation between mathematical modeling, state management, and visual rendering.

Architectural Block Diagram



Verified Technologies Utilized in the Project

Subsystem	Technology Used	Specific Functional Role
Frontend Core	React 19 · TypeScript · Vite 8	Type-safe component architecture, sub-second HMR, and modular 3-zone persistent layout shell.
State Management	Zustand (shallow slices)	Decoupled reactive telemetry store (simStore) managing live snapshots, profiles, and history.
Visualizations	Recharts · HTML5 Canvas	60 FPS waveform rendering, multi-point 7-day risk trend curves, and SVG radial gauges.
Deep Learning	EchoNext 1D ResNet-34	In-app convolutional neural network inference classifying simulated ECG rhythm and structural indices.
Document OCR	Tesseract.js (WASM)	In-browser optical character recognition extracting clinical lab values without backend dependency.
PDF Generation	jsPDF · HTML2Canvas	Client-side rendering and vector PDF compilation for downloadable clinical patient summaries.

Software Modularity: All calculations, risk weighting algorithms, and sensor mock pipelines reside in dedicated pure TypeScript modules (riskEngine.ts, hal/), allowing straightforward replacement with live hardware drivers.

Current Status and Limitations

Honest Engineering Assessment of Capabilities and Boundaries

In evaluating any clinical software system, clear boundaries between what is fully implemented and what is simulated or experimental are essential. Below is an honest breakdown of the platform's current maturity.

What is Currently Implemented

- Full 3-zone persistent clinical workstation interface.
- Synthetic physiological parameter generators for HR, BP, and HRV.
- Multi-algorithm composite risk scoring engine.
- EchoNext 1D ResNet-34 deep learning inference model.
- 7-Day longitudinal recorded risk history tracking.
- Client-side document OCR ingestion for PDF/image lab reports.
- Consolidated Patient Report generator with client-side PDF export.

Current Limitations & Boundaries

- **Hardware Disconnection:** Physical BLE/Bluetooth ECG/PPG sensor hardware drivers are currently mocked.
- **Simulation Heuristics:** Risk weighting heuristics have not undergone multi-center prospective clinical trials.
- **Regulatory Clearance:** Platform has not been reviewed or cleared by FDA, CE, or CDSCO.
- **Document Parsing Variability:** OCR accuracy depends on scan quality and varying commercial lab formats.
- **Model Scope:** Deep learning model is trained on simulated rhythm archetypes, not clinical 12-lead databases.

Hardware Dependency Roadmap

The hardware abstraction layer (ISensorSource) is designed with standardized interfaces. Transitioning from simulation to physical deployment requires replacing MockSensorSources.ts with real-time Bluetooth Low Energy (BLE) or serial GATT services streaming from certified patient monitors.

Prototype Standing: This software is strictly an academic, research, and engineering prototype designed to demonstrate multi-modal cardiovascular data integration. It is not approved for direct clinical diagnostic or therapeutic decision-making.

End-to-End Demo Workflow

Recommended Walkthrough Guide for Supervisors and Reviewers

To experience the platform's complete feature set during an evaluation or demonstration, follow this step-by-step clinical walkthrough:

Inspect Baseline State

- 1 Observe the default "Healthy — Baseline" state: Risk Score in the green band (approx. 19 Low), normal sinus rhythm, optimal BP (120/80 mmHg), and stable trend.

Click Arohan Logo to Preview Asset

- 2 Click the uncontained Arohan passport-size logo in the sidebar header to trigger the centered high-resolution modal preview; close via ESC or backdrop click.

Select Pathological Scenario

- 3 Open the header scenario dropdown and select "CAD — Significant Stenosis". Notice immediate updates across hemodynamics, elevated FAI (-65 HU), and rising CAC score.

Evaluate Composite Risk Shift

- 4 Observe the primary CAD Risk Score transition into the High-Risk range (65–75). Inspect the disease-specific risk bars (Atherosclerosis and Ischemia elevate).

Ingest External Clinical Data

- 5 Click "Upload Report" in the top-right header. Ingest a sample lab PDF/image, review extracted lipid parameters in the preview modal, and apply to the simulation.

Inspect Deep Learning Classifier

- 6 Examine the EchoNext ResNet-34 diagnostic card. Click "Re-Infer" to trigger a neural network pass and observe structural heart disease index updates.

Generate & Export Patient Report

- 7 Click "Patient Report" in the left sidebar. Review the synthesized clinical summary, XAI factor contributions, and click "Export PDF" to generate a clinical handover report.

Reviewer Conclusion: This 7-step sequence illustrates the complete data lifecycle—from scenario simulation and raw sensor processing to transparent risk stratification and downloadable clinical reporting.